

CHECK POINT CERTIFIED SECURITY ADMINISTRATOR

CCSA104 Practical Assignment
Managing Security Policies

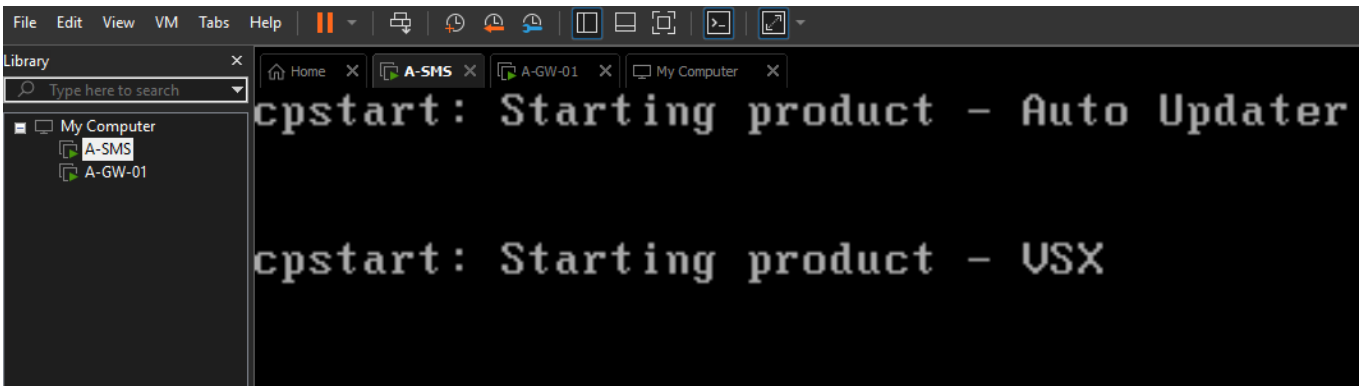
Report detail	Information
Student name	Cossam Kondowe
Student ID	C11/26/CCSA/17044
Course	CCSA104 Security Policy Management
Lab environment	R82 SmartConsole, A-SMS and Security Gateway environment
Security Management Server	A-SMS, 10.1.1.101

1. Scope, Safety and Brief Check

This report provides a step-by-step workflow for building, publishing, installing and verifying an Access Control policy for Orion Logistics. The design follows a Positive Control model: permitting only required traffic, drop everything else and log the result.

2. Environment Readiness

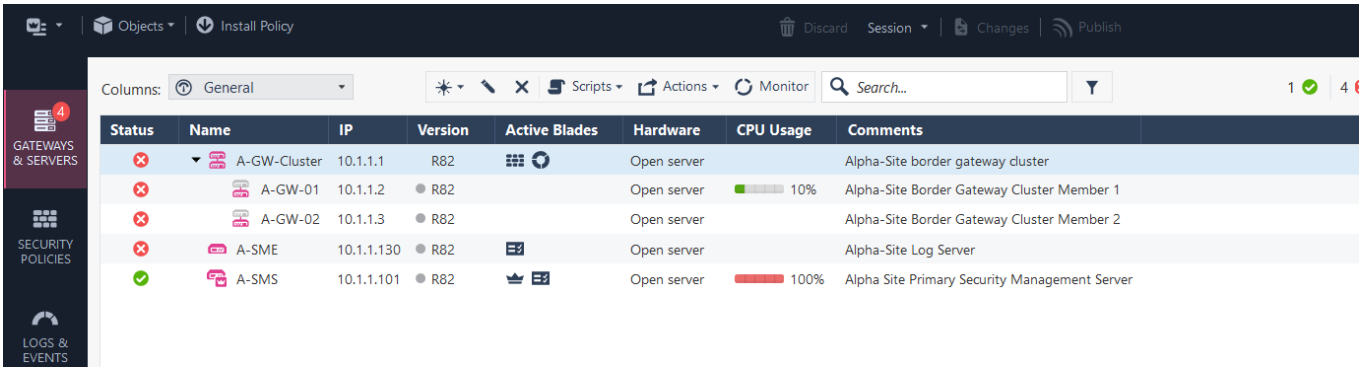
1. I powered A-SMS and the required gateway.



2. I open SmartConsole and connect to A-SMS at 10.1.1.101 using the supplied management administrator account.



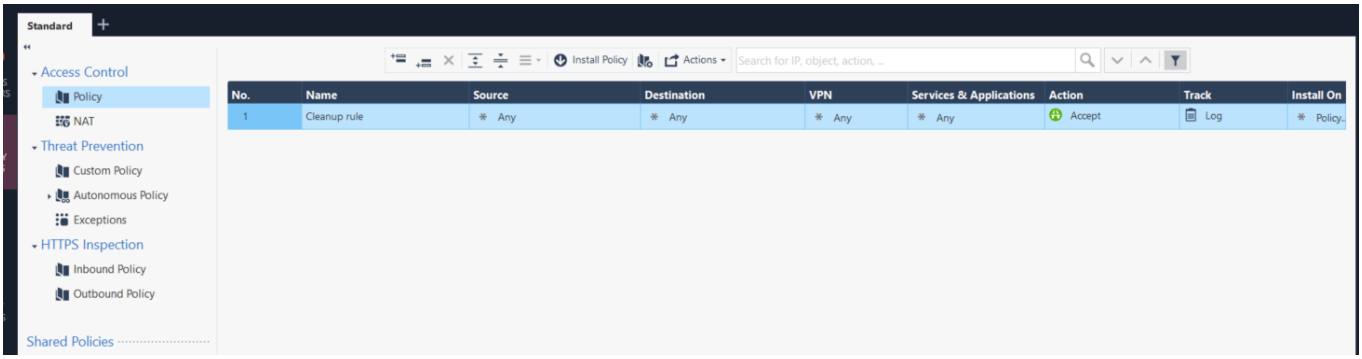
3. I open Gateways and Servers and confirm the intended gateway or cluster object.
4. Recorded the exact target object, member states, SIC/communication status and alerts.
5. Confirmed the Internal, DMZ and External VMware networks and test hosts from Lab 4A.
6. Did not reset SIC, delete cluster members or modify licences.

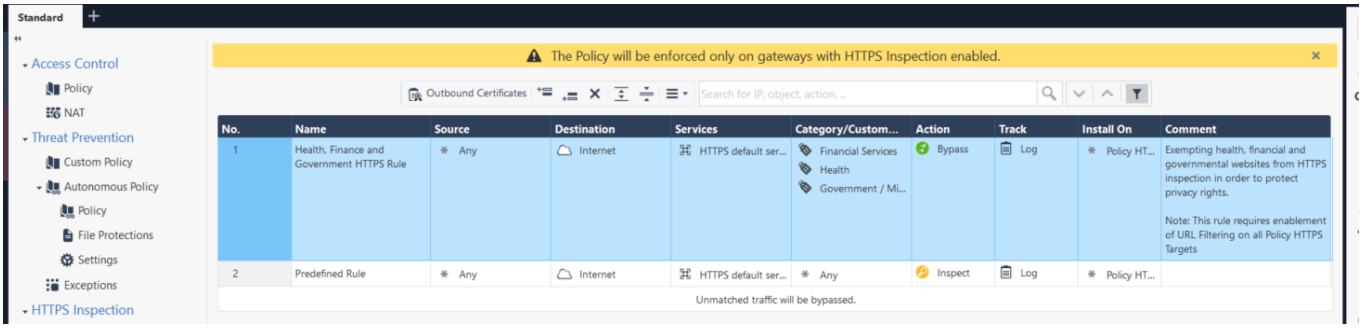


4. Part A: Practical Tasks

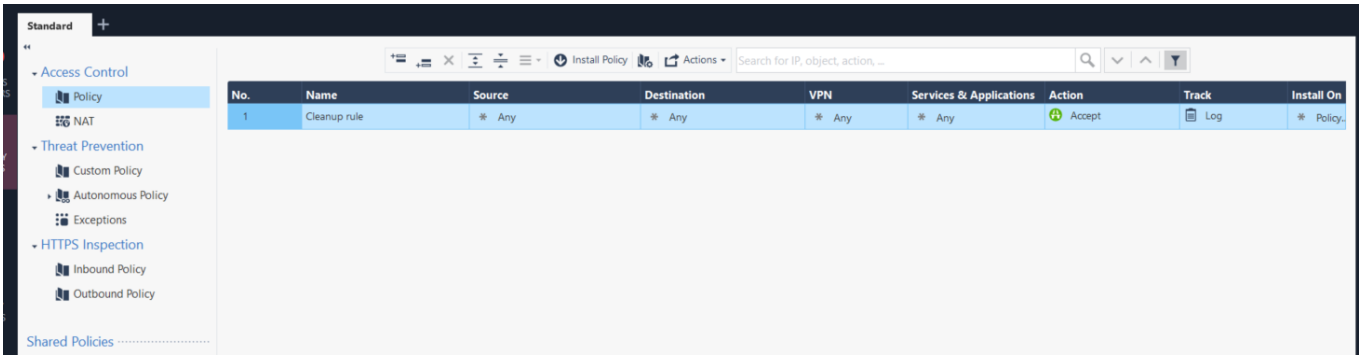
Task 1: Exploring the Default Policy

7. Selected Security Policies from the left navigation.
8. The active policy-package name.

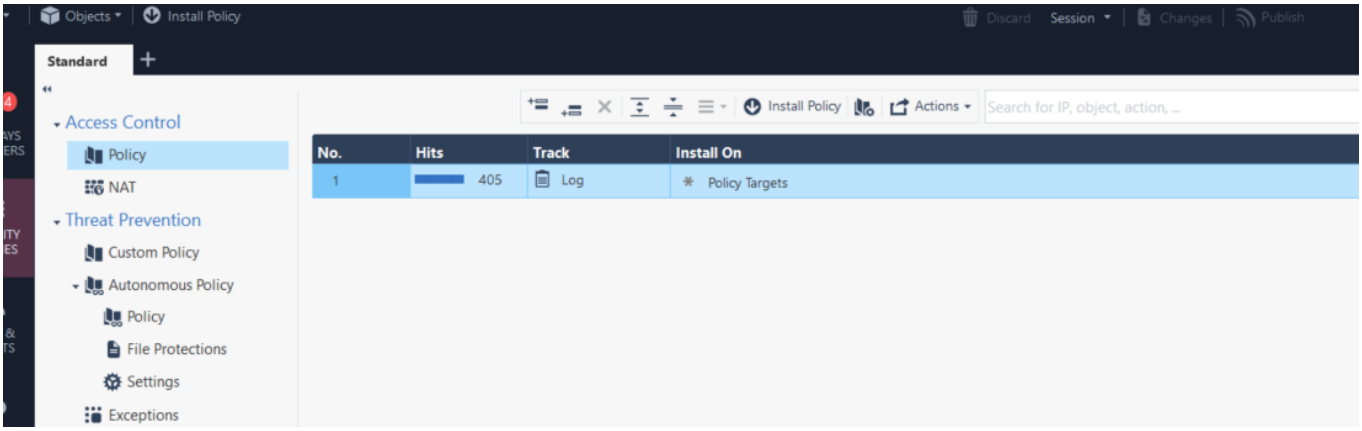




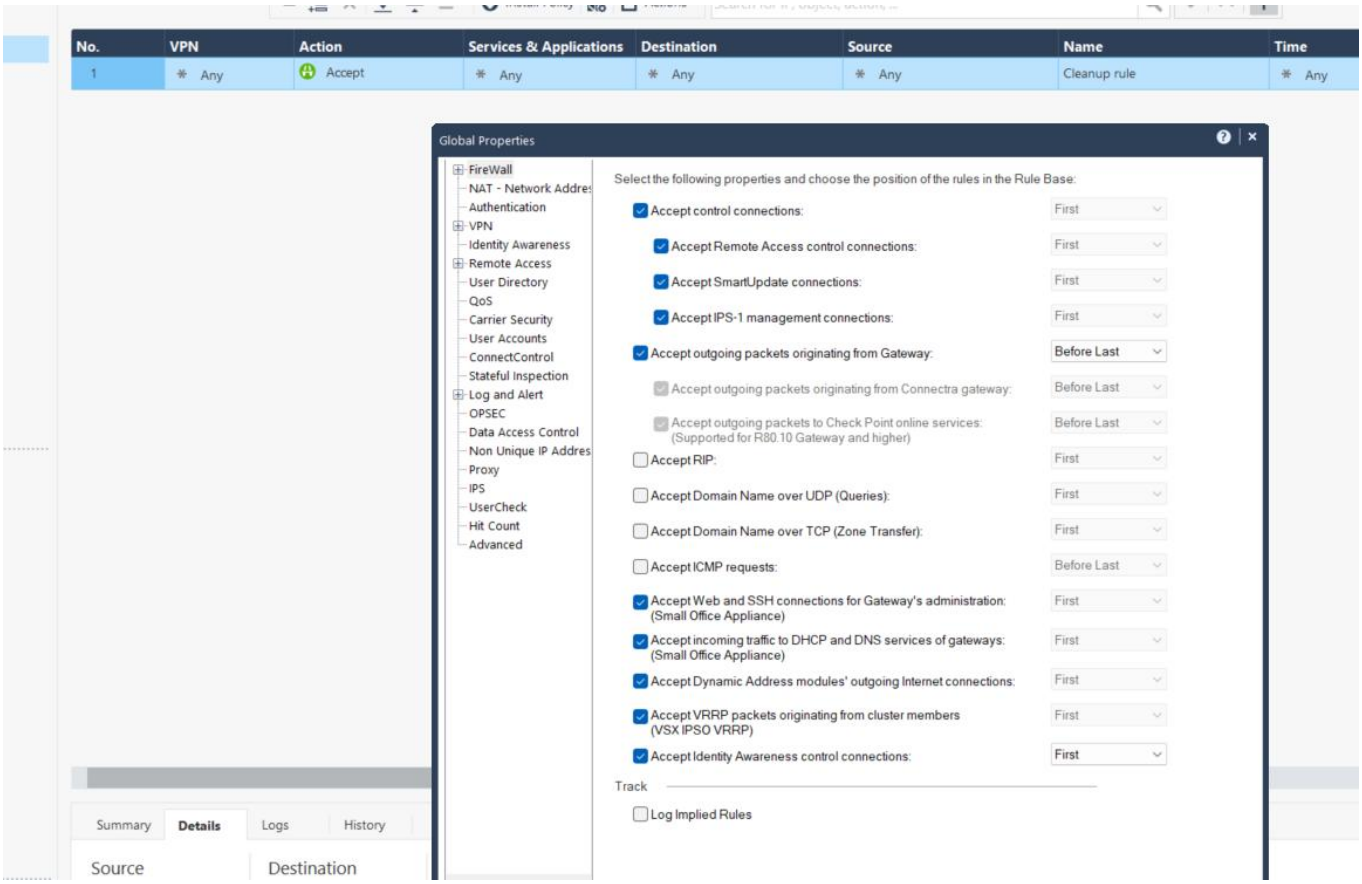
9. Recording the Cleanup rule fields exactly as currently shown.



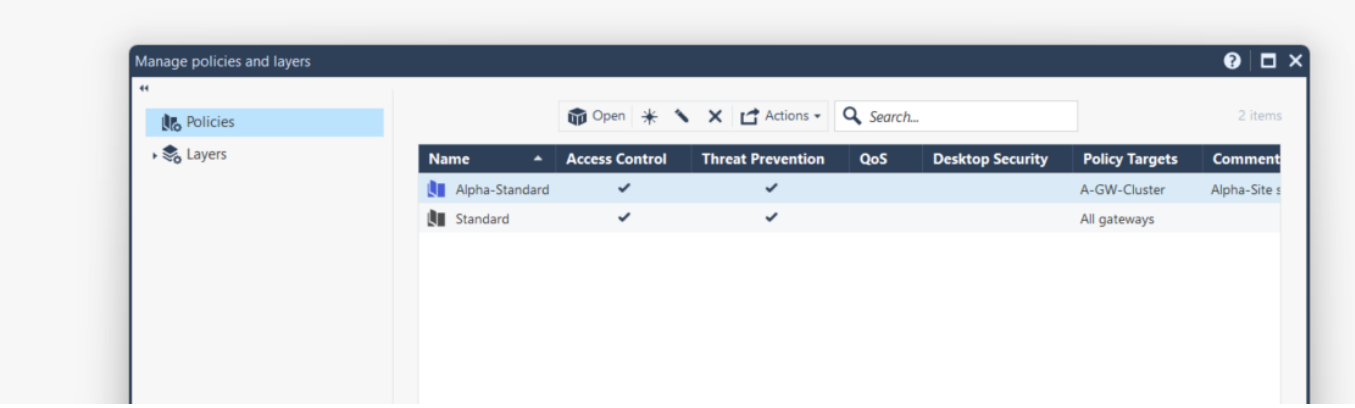
10. Enabling Hits, Track and Install On.

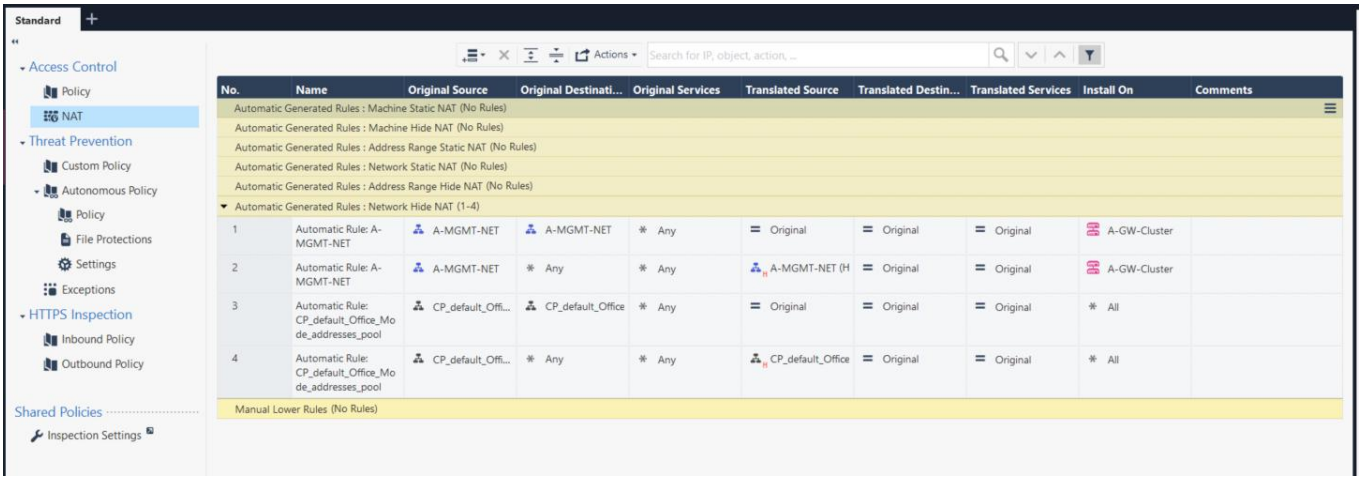


Global Properties. Recording the visible categories.



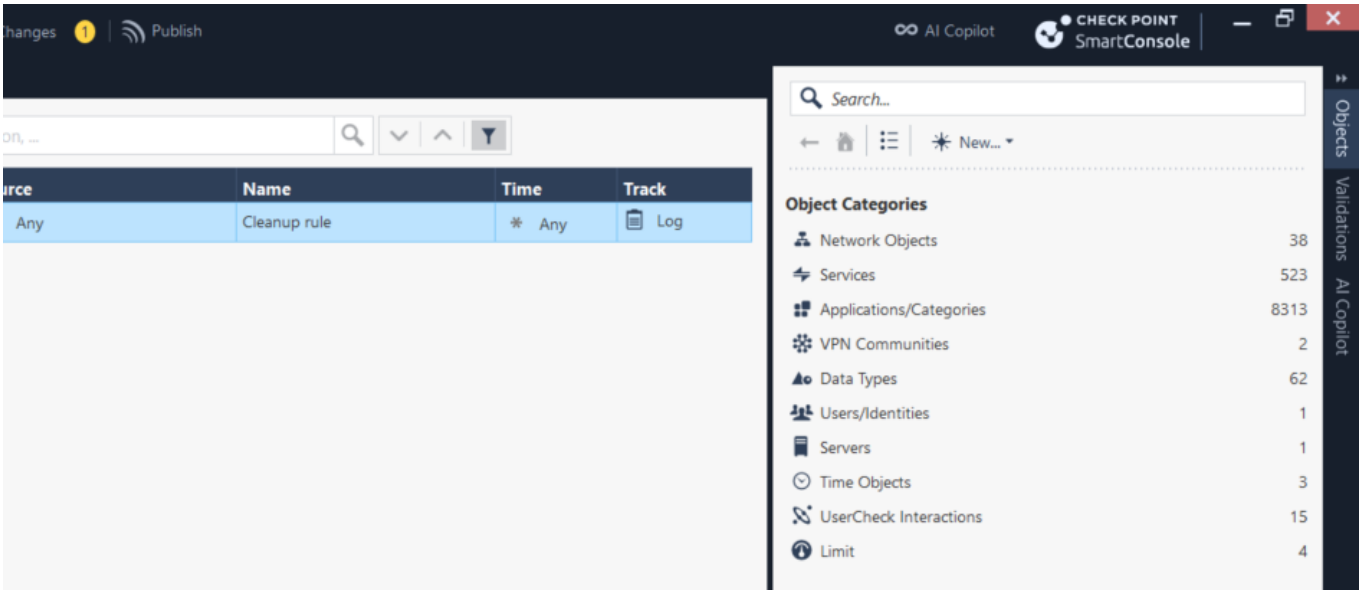
11. existing policy package.



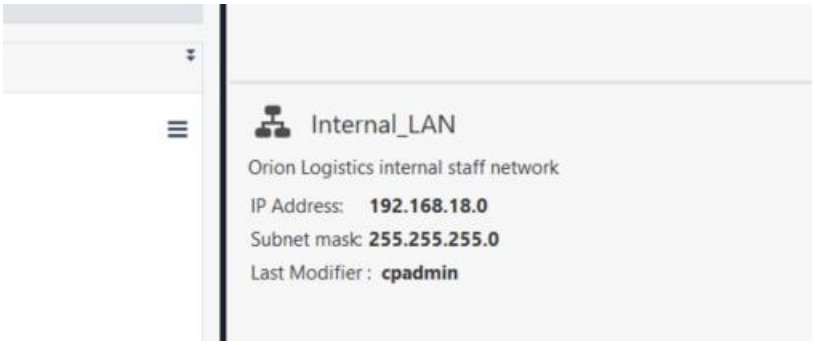


Task 2: Creating Objects and Build the Access Control Rulebase

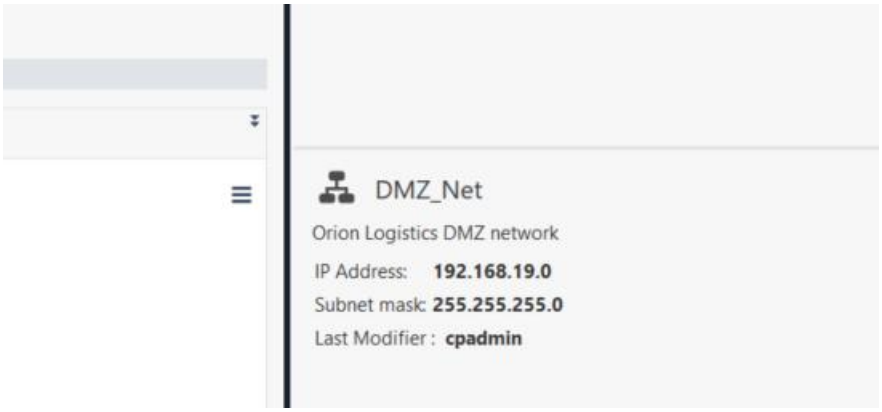
12. Open Objects > Object Explorer.



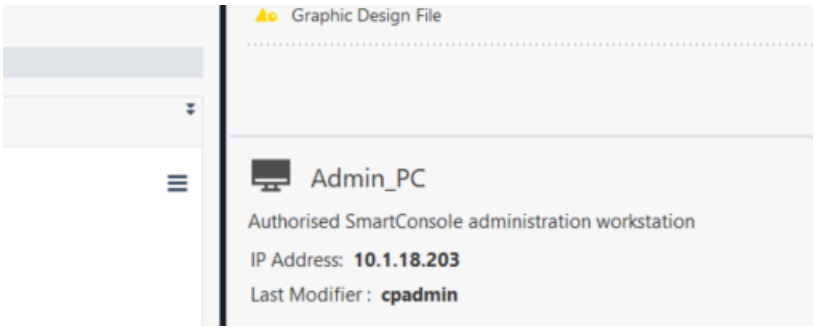
13. Internal LAN



DMZ_NET



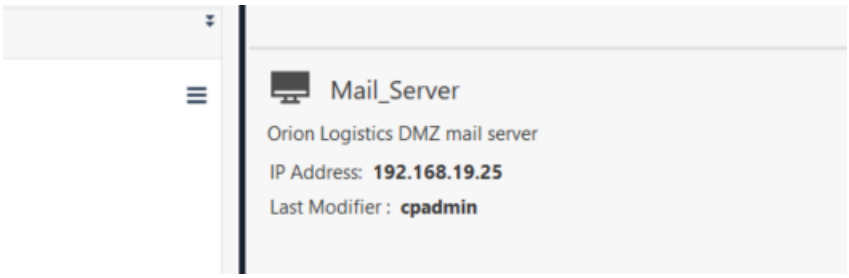
Admin_PC



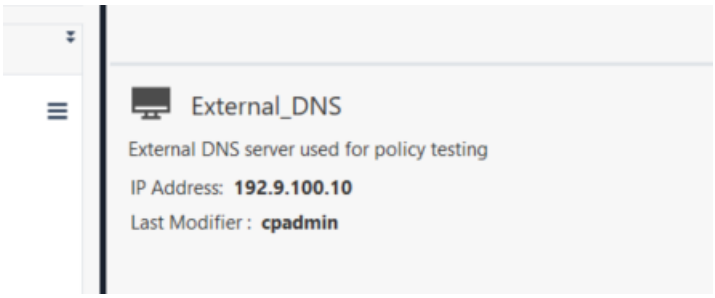
Web_Server



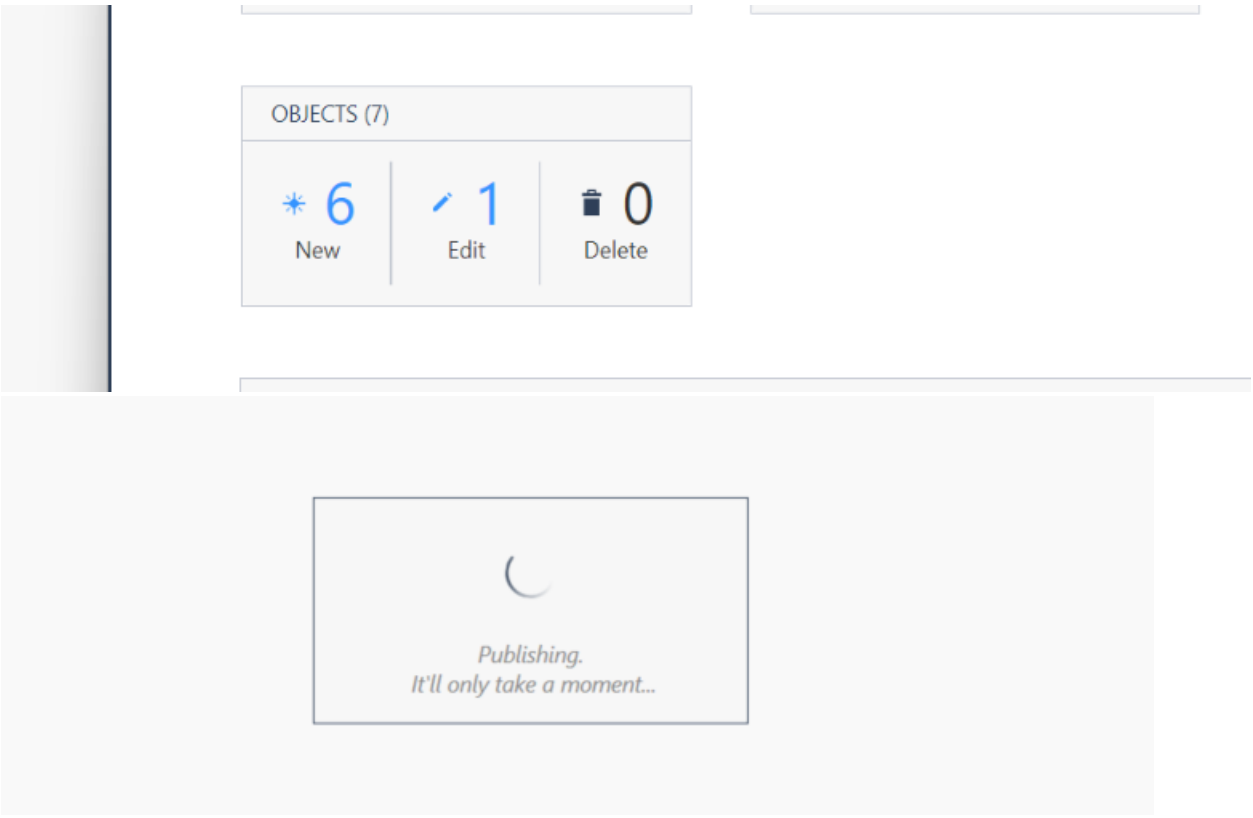
Mail_Server



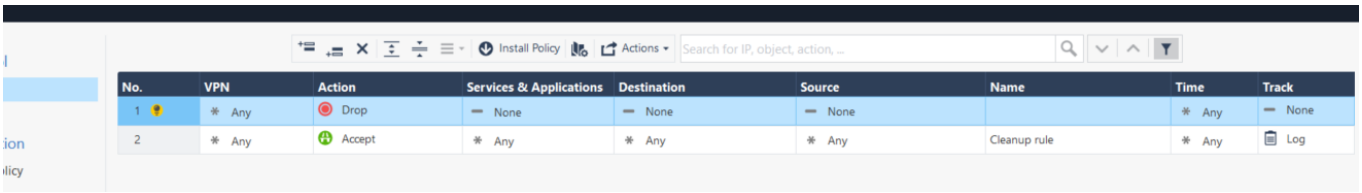
External_DNS



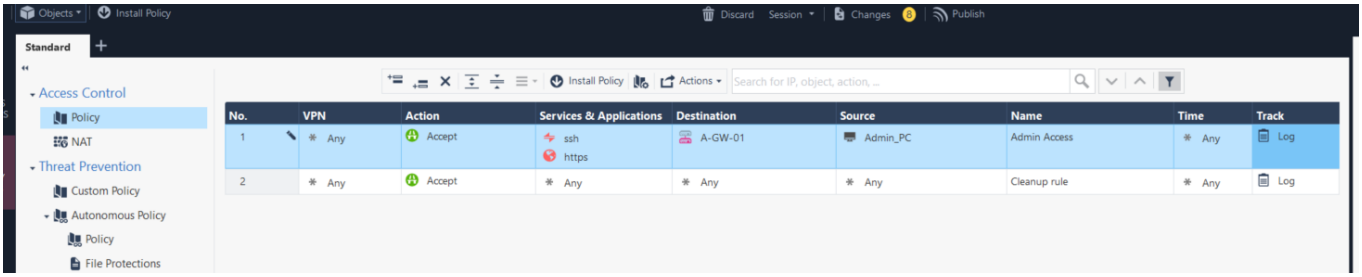
14. Publishing the object changes.



15. New rules above the existing Cleanup rule.

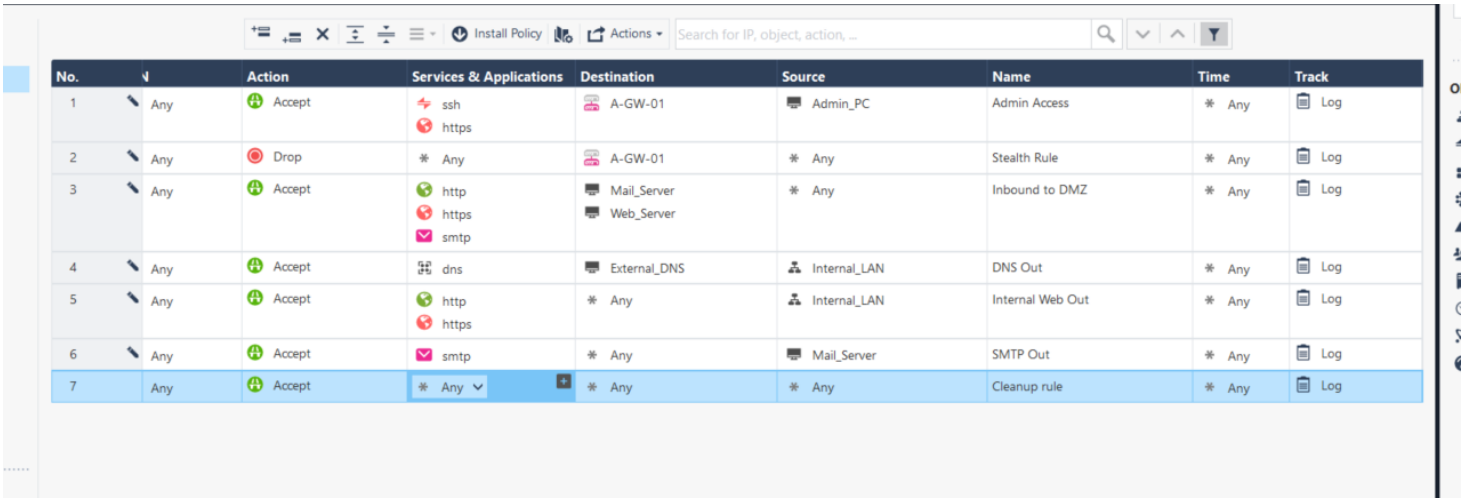


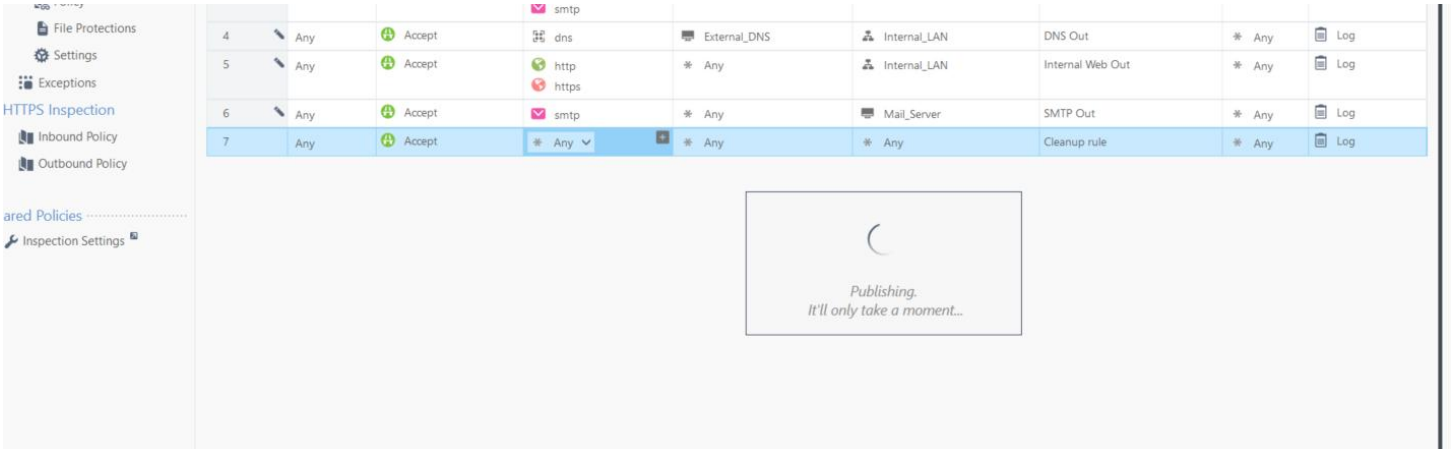
16. Use the exact rule names and fields shown below.



Pos.	Rule name	Source	Destination	Services	Action	Track
1	Admin Access	Admin_PC	Gateway	ssh, https	Accept	Log
2	Stealth Rule	Any	Gateway	Any	Drop	Log
3	Inbound to DMZ	Any	Web_Server, Mail_Server	http, https, smtp	Accept	Log
5	DNS Out	Internal_LAN	External_DNS	dns UDP and TCP	Accept	Log
6	Internal Web Out	Internal_LAN	Any	http, https	Accept	Log
7	SMTP Out	Mail_Server	Any	smtp	Accept	Log
8	Cleanup Rule	Any	Any	Any	Accept	Log

New created Rules





Task 4: Organising the Rulebase with Sections

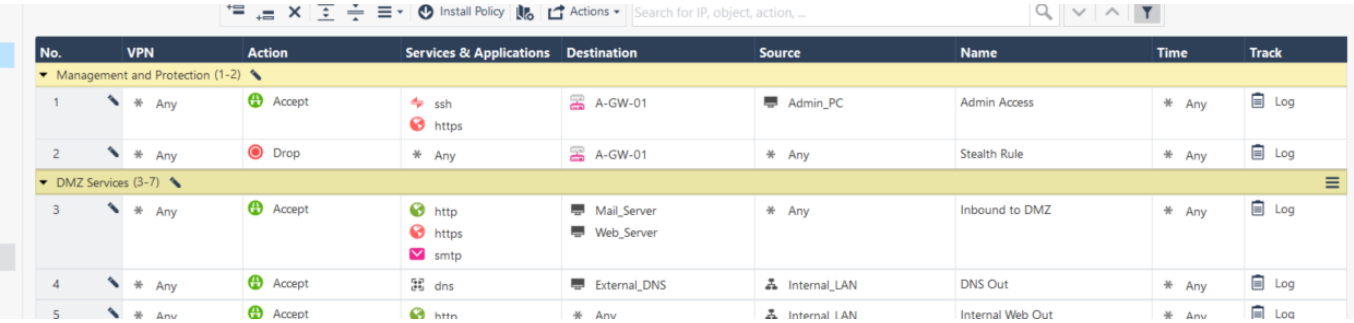
17. Section 1: Management and Protection

Admin Access

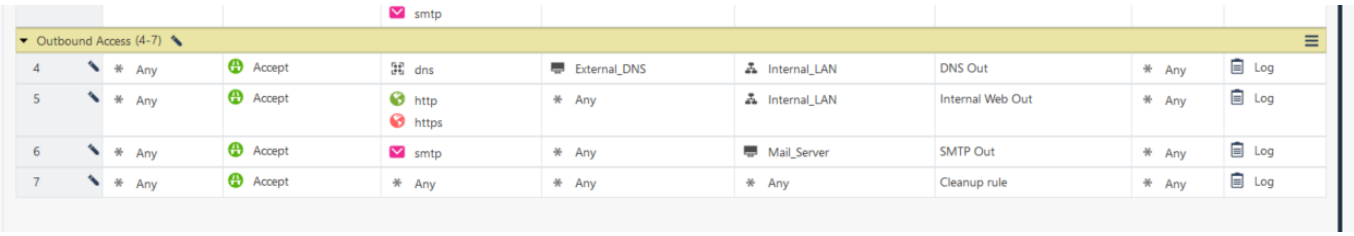
Stealth Rule



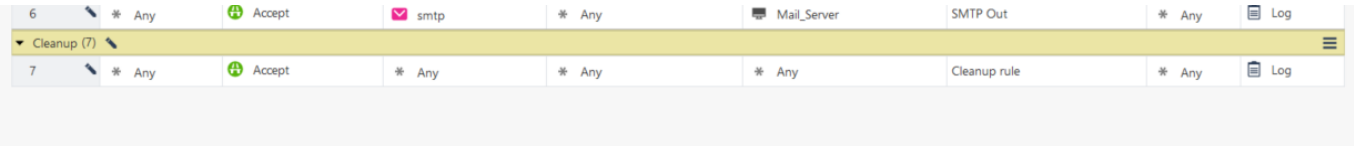
18. Section 2: DMZ Services



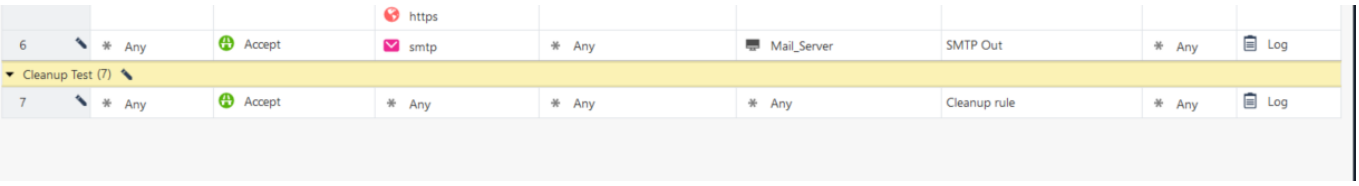
19. Section 3: Outbound Access



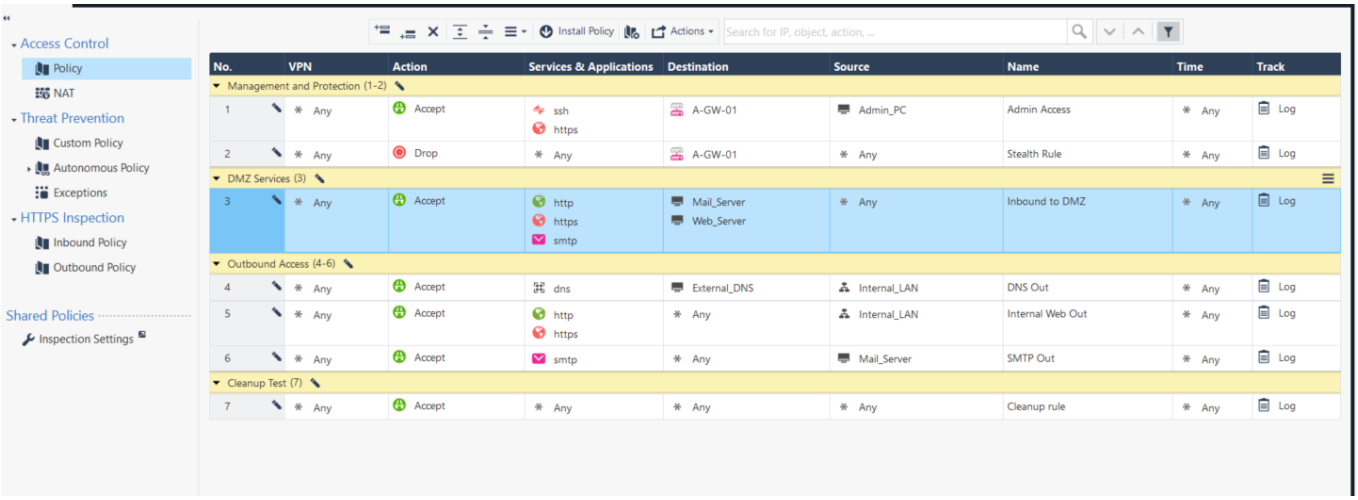
20. Section 4: Cleanup



21. Temporarily edit a section title

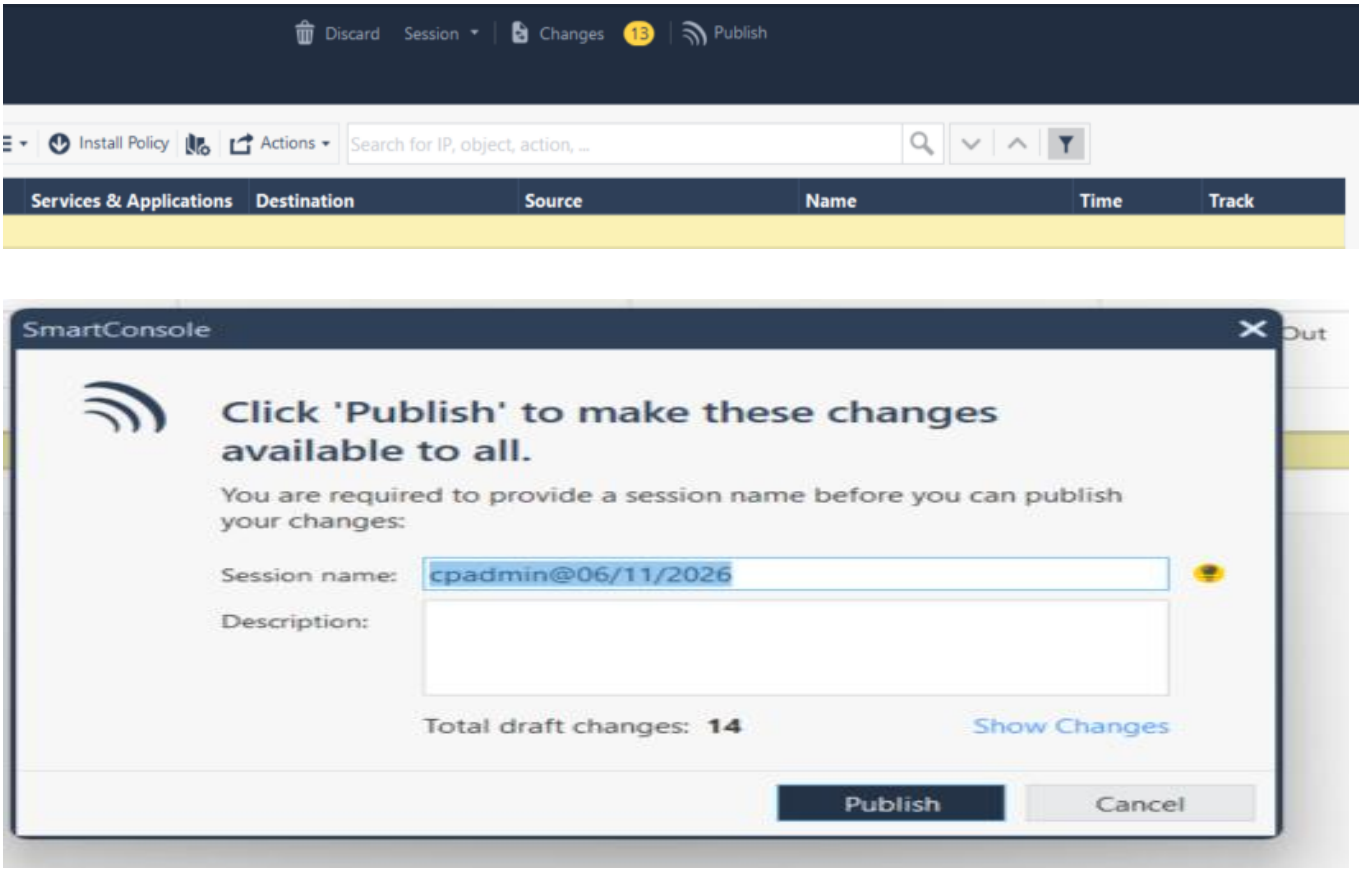


General Overview



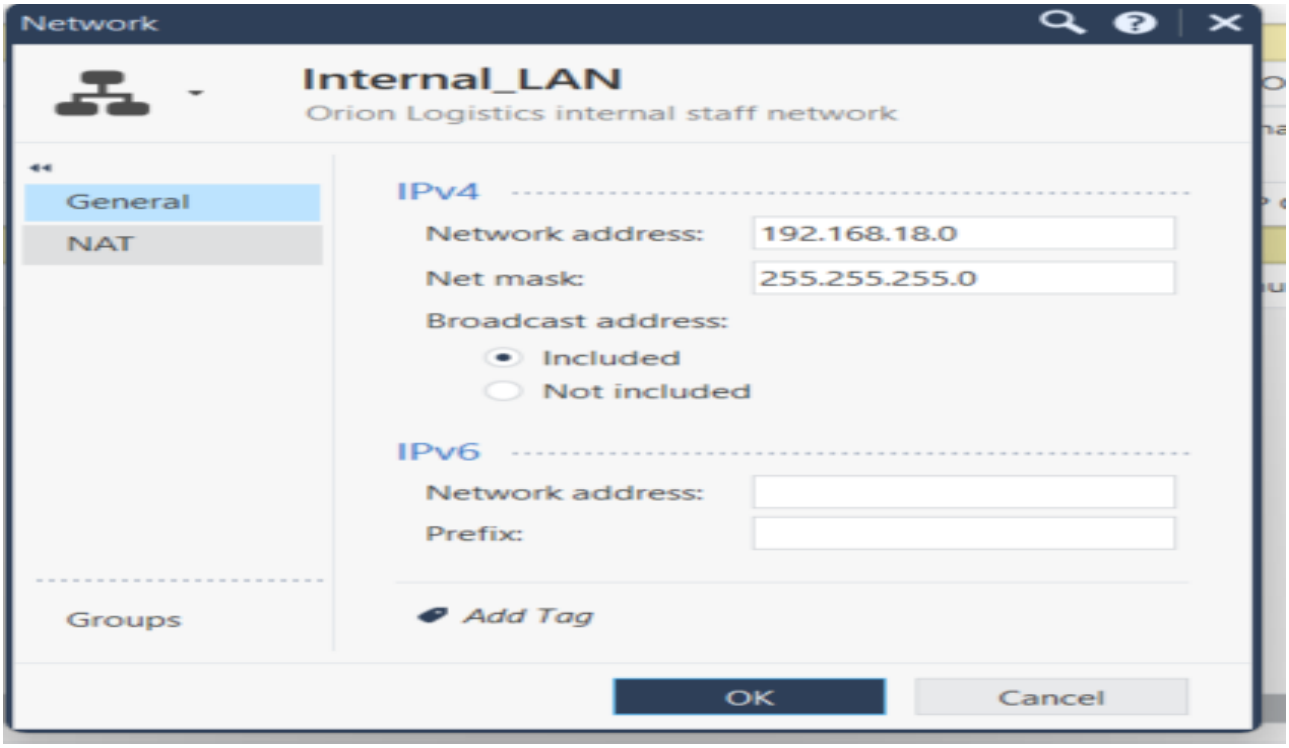
Rulebase sections are visual divisions used to organise related security rules and make the policy easier to read and maintain. Sections do not change the top-to-bottom enforcement order of the rules, and the section titles are not installed on the Security Gateway as enforcement rules. The gateway continues evaluating the actual Access Control rules according to their position in the rulebase.

Reviewing and publishing

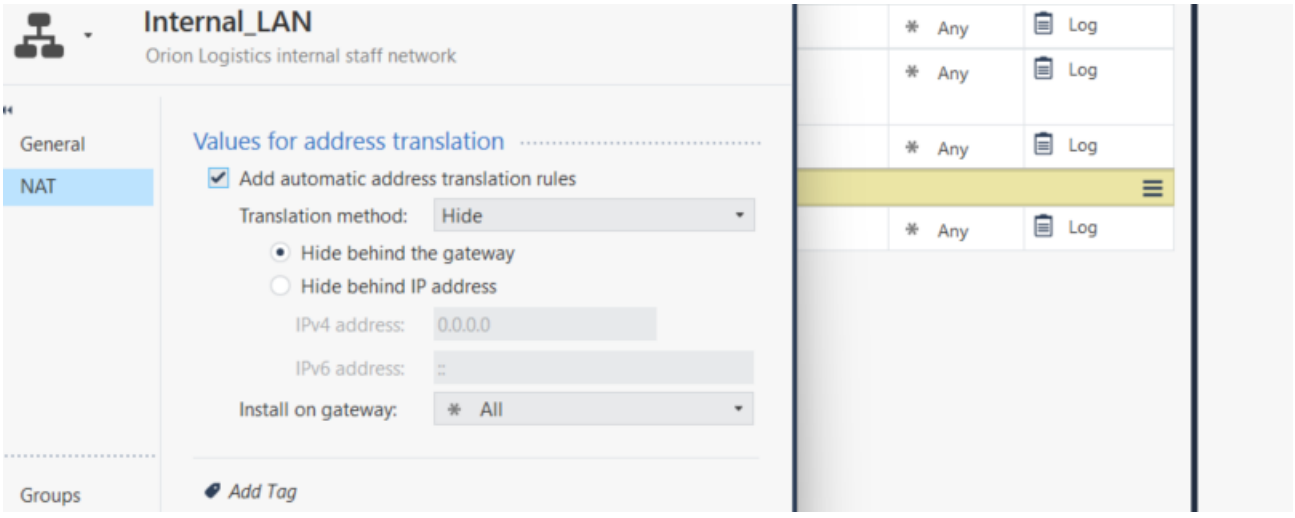


Task 6: Configuring Hide NAT

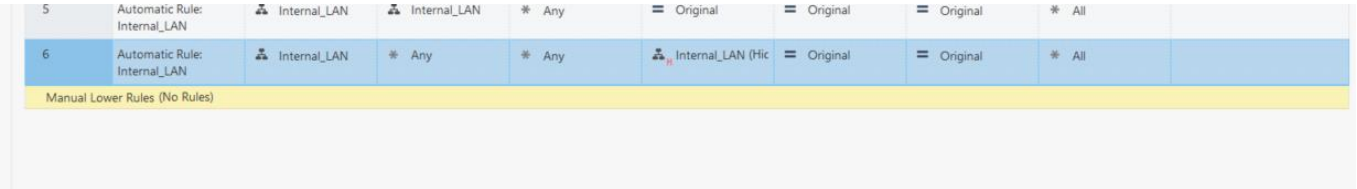
22. Editing Internal_LAN.



23. Open the NAT page.



24. Locating the automatic rule or rules generated for Internal_LAN.

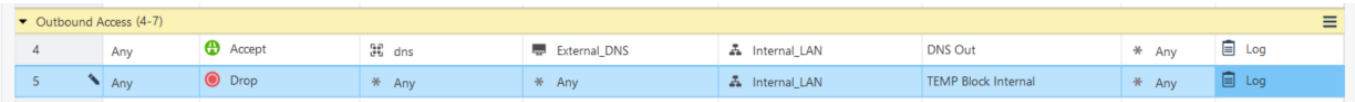


5	Automatic Rule: Internal_LAN	Internal_LAN	Internal_LAN	* Any	= Original	= Original	= Original	* All
6	Automatic Rule: Internal_LAN	Internal_LAN	* Any	* Any	Internal_LAN (Hic	= Original	= Original	* All
Manual Lower Rules (No Rules)								

This is an automatic NAT rule because SmartConsole generated it from the NAT settings of the Internal_LAN object instead of the administrator manually creating a row in the NAT rulebase. Hide NAT performs many-to-one translation, allowing many internal hosts to share the gateway's translated address. To keep simultaneous sessions distinct, the gateway can translate source ports and maintain a connection-state table that maps each translated session back to the correct internal host.

Task 7: Rule Order Experiment and Installation History

25. Creating TEMP Block Internal with Source Internal_LAN, Destination Any, Services Any, Action Drop and Track Log.



Check Point evaluates Access Control rules from top to bottom and stops at the first matching rule. When a broad drop rule is above a more specific allow rule, the broad rule can match first and prevent the intended allow rule from being reached. Specific business rules therefore belong above broader rules, while the most general Cleanup rule remains at the bottom.

Section B1 - Multiple Choice Questions (5 Marks)

Question	Correct Answer	Explanation
Q1	B	Created in SmartConsole, stored on the Security Management Server, enforced by Security Gateways.
Q2	A	Check Point Access Control uses a Positive Control Model (whitelist).
Q3	C	The Stealth rule should be placed as early as possible, typically immediately after the Management rules.
Q4	D	Rulebase sections are visual divisions only and do not affect enforcement order.
Q5	B	Hide NAT performs many-to-one translation

Section B2: Short Answers

Q6. Explicit rules, Implied rules and Implicit Cleanup

Explicit rules are the rules that an administrator deliberately creates and can see in the Access Control rulebase. Implied rules are generated from options in Global Properties, particularly the Firewall-1 Implied Rules settings, and can be positioned before the first rule, before the last rule or after the last rule depending on the selected option. The Implicit Cleanup Action is the default action applied when traffic reaches the end of a layer without matching an explicit or applicable implied rule; in a Positive Control design, unmatched traffic should be denied.

Q7. Why rule order matters

The gateway evaluates the rulebase from top to bottom and uses the first rule that matches the connection. In this rulebase, Admin Access must be above the Stealth Rule so that approved SSH and HTTPS management traffic can reach the gateway before the broader Stealth Rule drops other traffic addressed to the gateway.

Q8. Saving, publishing and installing

Saving keeps edits inside the current administrator session, while publishing commits the session changes to the shared management database so that other administrators can see and use them. Unpublished changes are not included in a normal policy installation because they have not been committed to the shared database. Many organisations publish and install during an approved change window so that review, rollback planning, user impact and audit records are controlled.

Q9. Global Properties and Policy Packages

Global Properties contains settings that can affect the whole management domain, such as implied firewall rules, stateful inspection behaviour or authentication-related defaults, and it is opened from the SmartConsole Applications menu. A Policy Package groups installable policy components for selected targets, such as Access Control and Threat Prevention layers, and packages are opened or managed through Security Policies and Manage policies and layers.

Q10. Security Zones

The four predefined Security Zones are InternalZone, ExternalZone, DMZZone and WirelessZone. A gateway interface can belong to only one zone at a time, and zone membership must follow the real gateway topology so that traffic direction is represented correctly. Zone-based rules improve readability and scalability because rules can refer to a logical trust area instead of listing every network attached to that area.

Section B3: Scenario Questions

Q11. Any-Any-Accept above the Stealth Rule

An Any-Any-Any Accept rule above the Stealth Rule would allow unauthorised traffic to reach the gateway itself and would also bypass the intended Positive Control design for other traffic because such a broad rule would match before the restrictive rules below it. The correction is to remove or narrow the rule to the exact approved sources, destinations and services, then keep the Stealth Rule immediately after authorised management rules. A specific management-access rule is allowed above the Stealth Rule because authorised administrators must be permitted to reach the gateway before the broader protection rule drops all other direct access.

Q12. Hide NAT troubleshooting

(a) NAT conserves public IPv4 addresses by allowing multiple internal systems to share a translated address, and it hides internal addressing from external networks.

(b) First, I would verify the Internal_LAN object and its automatic Hide NAT settings, including the network/mask and the selected hide-behind gateway, because a wrong object or target produces an incorrect translated rule. Second, I would confirm that the required Access Control rule permits the outbound traffic and that the relevant NAT and Access Control policy was published and successfully installed on the correct gateway or cluster. Third, I would check routing and connectivity: the gateway must have a working external interface and default route, the client must use the correct gateway, and the return path must lead back through the same Check Point gateway. Logs and the policy installation result would show whether the failure is caused by policy, translation or routing.

Limitations

The supplied CCSA104 Practical Assignment document did not include the instructions for Task 3 and Task 5. The document moved directly from Task 2 to Task 4, and from Task 4 to Task 6. As a result, the required procedures, configuration values and evidence requirements for these two tasks were unavailable.

Task 3 was not completed because the supplied assignment did not explain the activity to be performed, the expected outcome or the screenshot evidence required. Completing the task without official instructions would have involved making assumptions that could affect the accuracy of the configuration and assessment evidence.

Task 5 was also not completed because its Security Zone assignments and the required fields for the Internal to DMZ rule were not provided. Although the assignment indicated that Task 5 would add this rule, it did not specify the source, destination, services, action, tracking option, interface-to-zone mappings or required evidence.

Therefore, no screenshots or results have been included for Tasks 3 and 5. The omission has been recorded honestly, and no evidence has been fabricated. All other tasks for which clear instructions were available were completed and documented using evidence from the lab environment

9. References

CCSA104 Practical Assignment: Managing Security Policies.

Companion Lab 4A: Manage Security Policies.

Check Point R82 SmartConsole and Security Management course materials.